M378K : November 6th, 2026.

t-Distribution.

Def'n. A **Student t-Distribution** w/ k degrees of freedom is the distn of the r.v.

$$T = \frac{Z}{\sqrt{\frac{Q^2}{k}}}$$

w/ • $Z \sim N(0,1)$

• $Q^2 \sim \chi^2(df = k)$

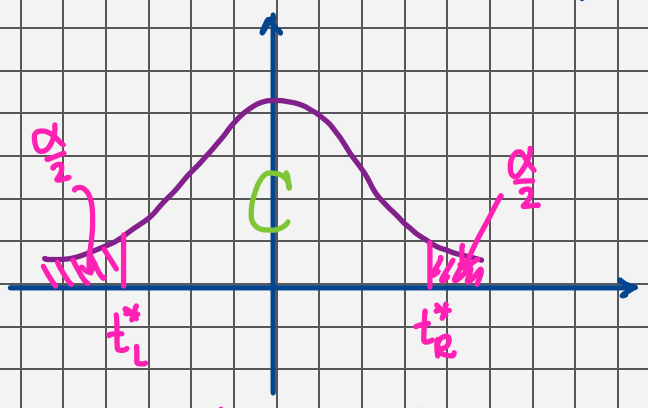
• Z and Q^2 are independent

We write $\boxed{T \sim t(df = k)}$

$$S^2 = \frac{1}{\cancel{n-1}\ n} \sum_{i=1}^{n} (Y_i - \cancel{\bar{Y}}\ \mu)^2 \quad \checkmark$$

Back to Confidence Intervals

To construct a CI w/ $C = 1-\alpha$, we do



$$t^* = t_R^* = -t_L^*$$

$$t^* = qt(1-\alpha/2, df = n-1)$$

$$t^* = qt((1+C)/2, df = n-1)$$

$$P\left[-t^* \le \frac{\bar{Y}-\mu}{\frac{S}{\sqrt{n}}} \le t^*\right] = C$$

$$P\left[-t^* \cdot \frac{S}{\sqrt{n}} \le \bar{Y}-\mu \le t^* \cdot \frac{S}{\sqrt{n}}\right] = C$$

$$P\left[\underbrace{\bar{Y}-t^* \cdot \frac{S}{\sqrt{n}}}_{=\hat{\theta}_L} \le \mu \le \underbrace{\bar{Y}+t^* \cdot \frac{S}{\sqrt{n}}}_{=\hat{\theta}_R}\right] = C = 1-\alpha$$

$$\mu = \bar{Y} \pm t^* \cdot \frac{S}{\sqrt{n}}$$

$\bar{Y}$: pt. estimator; $\frac{S}{\sqrt{n}}$: std error; t^*: critical value

Problem 16.9. *(20 points)*
A random sample of size 10 *is drawn from a normal distribution with* **both** *mean and standard deviation* **unknown.** *The generated sample has the sample mean* $\bar{y}_{10} = 14$ *and the (unbiased) estimate of the variance* $s^2 = 25$.

(i) (10 points) Construct a (symmetric) 90%-confidence interval for μ.

(ii) (10 points) Construct a (symmetric) 90%-confidence interval for σ^2.
Hint: *Remember that you know the distribution of* $(n-1)S^2/\sigma^2$.

$\alpha = 0.10$

i.

$$\mu = \bar{Y} \pm t^* \cdot \frac{S}{\sqrt{n}}$$

$t^* = qt(0.95, df=9)$

$$\mu = 14 \pm t^* \cdot \frac{5}{\sqrt{10}}$$

t^*: 1.833 (from the t-tables)

ii.

$$Q^2 = \frac{(n-1)S^2}{\sigma^2} \sim \chi^2(df=9)$$

$a = qchisq(0.05, df=9)$

$b = qchisq(0.95, df=9)$

$$\mathbb{P}\left[\frac{(n-1)S^2}{b} \leq \sigma^2 \leq \frac{(n-1)S^2}{a}\right] = 0.90$$

$$= \mathbb{P}\left[a \leq \frac{(n-1)S^2}{\sigma^2} \leq b\right] = 0.90$$

$$CI = \left(\frac{9 \cdot 25}{b}, \frac{9 \cdot 25}{a}\right)$$

b: 16.92, a: 3.325

M378K Introduction to Mathematical Statistics
Problem Set #17
Relative efficiency.

Definition 17.1. *Given two unbiased estimators $\hat{\theta}_1$ and $\hat{\theta}_2$, the efficiency of $\hat{\theta}_1$ relative to $\hat{\theta}_2$ is defined as*

$$eff(\hat{\theta}_1, \hat{\theta}_2) = \frac{\mathrm{Var}[\hat{\theta}_2]}{\mathrm{Var}[\hat{\theta}_1]}.$$

Problem 17.1. *Let Y_1, Y_2 be a random sample from the exponential distribution with the unknown parameter θ.*

(i) *The estimator $\hat{\theta}_1 = (Y_1 + Y_2)/2$ for θ is proposed. What is its variance?* ✓

(ii) *The estimator $\hat{\theta}_2 = cY_{(1)}$ for θ is proposed. Find the constant c such that $\hat{\theta}_2$ is an unbiased estimator of θ. What is its variance?* ✓ ✓

(iii) *Calculate the efficiency of $\hat{\theta}_1$ relative to $\hat{\theta}_2$.*

→ i. $\mathrm{Var}[\hat{\theta}_1] = \mathrm{Var}\left[\frac{1}{2}(Y_1+Y_2)\right] = \frac{1}{4}(\mathrm{Var}[Y_1]+\mathrm{Var}[Y_2])$

$= \frac{1}{2}\mathrm{Var}[Y_1] = \frac{\theta^2}{2}$

ii. $Y_{(1)} \sim E(\theta/2)$

$E[\hat{\theta}_2] \stackrel{?}{=} \theta$

$c \cdot E[Y_{(1)}] \stackrel{?}{=} \theta$

$c \cdot \frac{\theta}{2} = \theta \Rightarrow \boxed{c = 2}$

$\mathrm{Var}[\hat{\theta}_2] = \mathrm{Var}[2 \cdot Y_{(1)}] = 4 \cdot \mathrm{Var}[Y_{(1)}] = 4 \cdot \frac{\theta^2}{4} = \theta^2$

iii. $eff(\hat{\theta}_1, \hat{\theta}_2) = \frac{\mathrm{Var}[\hat{\theta}_2]}{\mathrm{Var}[\hat{\theta}_1]} = \frac{\theta^2}{\frac{\theta^2}{2}} = 2$ □